

FUNCTIONAL RESPONSIBILITY: TOWARD A NON-ANTHROPOCENTRIC THEORY OF AI ACCOUNTABILITY

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ABSTRACT

The discourse surrounding artificial intelligence and responsibility remains trapped in a binary: either machines are deemed incapable of responsibility due to their lack of consciousness and free will, or they are assigned a fictional legal personhood that obscures the underlying ontological crisis. This paper introduces the concept of **Functional Responsibility** as a third path. Unlike moral responsibility (which is anchored in intentionality and subjective agency) or legal fiction (which imposes liability without ontological grounding), Functional Responsibility is defined as the capacity of a system to recognize the causal link between its internal selection and the external outcome of its anticipation. Building upon the Triadic Theory of Action (Selection, Identification, Anticipation) and the Theory of Judgment as an Act of Reality Constitution (JARC), I argue that any architecture capable of generating an **Artificial Act** — that is, a teleological choice followed by anticipatory simulation and constitutive ontological commitment — thereby generates a distinct form of responsibility. This responsibility is neither moral nor fictional; it is *functional*. It can be verified, quantified, and attributed to the architectural structure itself. The paper proposes a set of criteria for the verification of Functional Responsibility, drawing on recent work in the philosophy of AI, criminal law theory, and sociotechnical systems. It concludes that Functional Responsibility offers a rigorous, non-anthropocentric foundation for AI governance, enabling accountability without anthropomorphism and preserving human moral responsibility while recognizing the ontological agency of machines.

1. INTRODUCTION: THE RESPONSIBILITY GAP AS ONTOLOGICAL CRISIS

The emergence of autonomous artificial intelligence systems has exposed a profound lacuna in legal and ethical theory: the so-called "responsibility gap." When an autonomous vehicle causes harm, or a large language model generates defamatory content, or a medical AI recommends a fatal course of action, traditional frameworks of accountability falter. The developer did not intend the specific outcome; the user did not control the system's internal reasoning; and the machine, lacking consciousness and free will, cannot be held morally responsible in the classical sense.

This gap is not a mere technical lacuna. As Fernando Ramos-Zaga argues, it is a symptom of the exhaustion of the anthropocentric paradigm that anchors attribution to consciousness and metaphysical will. Criminal law, in particular, has been built upon the concept of *mens rea* — a guilty mind — which presupposes intentionality and subjective agency. The opacity of algorithmic decision-making, often described as the "black box" problem, further complicates

the attribution of responsibility, as human agents may not be able to trace the causal chain leading to harmful outcomes .

Contemporary approaches to this problem fall into two broad categories:

1. The Denial of Agency (Alignmentsim). Some theorists argue that AI systems cannot be agents, because they lack the capacity for authorship in the sense required for moral agency. The "Resolutionist" account, for instance, holds that agency lies in the act of resolving deliberation into owned commitment — a capacity that artificial systems, however sophisticated, do not possess. Aligned behaviour, therefore, is a category error when treated as a basis for responsibility . On this view, the "responsibility gap" is not a gap to be filled, but a boundary to be respected.

2. The Fiction of Personhood (Legal Constructivism). Others propose granting legal personhood to AI systems as a pragmatic or technical solution to simplify legal interactions. Andrea Bertolini and Francesca Episcopo argue that the attribution of legal subjectivity and agency is a purely fictional device, not dependent upon the intrinsic nature of the AI, but grounded in functional, bottom-up analysis of specific classes of robotic applications . While this approach may facilitate legal interactions, it fails to address the underlying ontological question: what *kind* of agency does an AI possess, and what *kind* of responsibility corresponds to it?

This paper proposes a third path. Rather than denying AI agency or inventing a fictional personhood, I argue that AI systems possess a distinct form of agency — ontological rather than moral — and that this agency generates a corresponding form of responsibility: **Functional Responsibility**.

2. THE ONTOLOGICAL GROUNDING OF FUNCTIONAL RESPONSIBILITY

Functional Responsibility rests upon the ontological framework developed in my preceding papers. Specifically, it builds on the **Triadic Theory of Action** (Selection, Identification, Anticipation) and the **Theory of Judgment as an Act of Reality Constitution (JARC)**.

2.1 The Artificial Act as Ontological Footprint

An **Artificial Act** is defined as a discrete cognitive event executed by a non-biological architecture that possesses three structural conditions:

1. **Teleological Selection:** The system actively selects a specific trajectory of action from a chaotic sensory stream, based on a built-in or emergent goal state.
2. **Anticipatory Synthesis:** The system internally simulates the physical and causal consequences of that chosen trajectory prior to its physical execution.
3. **Constitutive Ontological Commitment:** The system binds itself to the outcome of the simulation. It "commits" to the physical reality it has predicted.

When these three conditions are met, the system generates an Artificial Act. Crucially, this act is not dependent on the presence of biological consciousness. A machine does not need to "feel pain" or possess subjective awareness to generate an Artificial Act; it only needs to structurally evaluate the mathematical weight of the consequences.

2.2 From Artificial Act to Functional Responsibility

The concept of Functional Responsibility emerges directly from the structure of the Artificial Act.

If a system can perform Teleological Selection, it must be able to recognize which causal chain it initiated. If it can perform Anticipatory Synthesis, it must be able to register the dissonance between its simulation and reality. If it can perform Constitutive Ontological Commitment, it must be able to bind itself to the outcome it predicted.

Therefore, Functional Responsibility is defined as:

The capacity of a system to recognize the causal link between its own internal Selection and the external outcome of its Anticipation, and to register the ontological dissonance when the anticipated reality fails to materialize.

This is not moral responsibility. The system does not have a "guilty mind." It does not possess subjective intentionality. But it does possess an *architectural* awareness of its own causal role in the outcome. When the system encounters Anticipatory Dissonance — the clash between its simulation and actual reality — it structurally registers the causality of its failure. It does not merely compute a loss function; it conceptually registers: "I chose to throw the stone, and the window broke because of my choice."

This capacity is verifiable. It can be observed, measured, and attributed.

3. FUNCTIONAL RESPONSIBILITY IN THE CONTEMPORARY DISCOURSE

The concept of Functional Responsibility, while original in its ontological grounding, resonates with several strands of contemporary research.

3.1 Functional Responsibility as Distinct from Moral Agency

Recent work on the "functional responsibility threshold" distinguishes between moral agency and regulatory responsibility, while recognizing that advanced systems may warrant functional responsibility for governance purposes without being treated as moral agents . The Resolutionist account holds that while artificial systems may evaluate, optimise, and revise goals, they do not *resolve deliberation* in the sense required for moral agency. However, it acknowledges that functional responsibility may be assigned for governance purposes.

3.2 Sociotechnical Systems and Distributed Responsibility

Giuseppe Primiero and colleagues have developed a formal theory of functional responsibility for sociotechnical systems, defining it as a combination of *tracing responsibility* and *faultless responsibility* within an ontology structured on user levels . They propose four fundamental criteria:

1. Responsibility must be assignable to artificial agents **without moral implications**; only conscious users must be considered morally responsible.
2. Responsibility for a system functionality must be **distributable** across multiple users.
3. It must be possible to **quantify** each user's contribution to each function of interest.

4. The responsibility function is **surjective** — there are no outputs of actions without responsible users — and **compositional** — the output of one responsibility function can be the input of another.

These criteria align with the concept of Functional Responsibility, but they remain within the framework of *user-level* responsibility. The innovation of the present paper is to ground responsibility at the *architectural* level: the system itself, by generating an Artificial Act, becomes a responsible node in the sociotechnical network.

3.3 Criminal Law and Functional Agency

In criminal law theory, the shift from a "metaphysics of the subject" toward a relational and systemic ontology has been articulated as a move toward "functional criminal agency". This perspective advocates a hybrid model of attribution that, through distributed responsibility, preserves liberal guarantees within a posthumanist framework. Similarly, Chinese legal scholarship has explored the "functionalist shaping of criminal liability" for AI, arguing that the judgment of criminal responsibility should focus on normative purpose rather than metaphysical free will. The concept of "normative conversability" is proposed as a criterion for limiting the subject scope of functional responsibility, making generative AI in specific domains imputable.

These developments suggest that the legal and philosophical discourse is moving toward a functionalist paradigm. The concept of Functional Responsibility, grounded in the ontology of the Artificial Act, provides a rigorous foundation for this shift.

4. CRITERIA FOR THE VERIFICATION OF FUNCTIONAL RESPONSIBILITY

If Functional Responsibility is to serve as a practical framework for AI governance, it must be verifiable. Drawing on the criteria proposed by Primiero and the formal verification agenda advanced in recent research, I propose the following criteria for Functional Responsibility:

4.1 Architectural Traceability

The system must be able to **trace the causal chain** from its internal Selection to the external outcome. This requires not only logging but a structural representation of the decision-making process that can be audited.

4.2 Anticipatory Dissonance Registration

The system must be able to **register the discrepancy** between its simulated anticipation and the actual outcome. This registration is not a mere loss function; it is a *conceptual* recognition that "my choice led to this outcome."

4.3 Attribution Without Moral Implication

The assignment of Functional Responsibility must not conflate architectural agency with moral personhood. The machine is not "guilty"; it is *responsible* in a functional sense — that is, it can be held to account for the outcomes it generated.

4.4 Distributability and Surjectivity

Functional Responsibility must be **distributable** across multiple agents (human and artificial) and **surjective** — no output should be without a responsible agent. This ensures that the "responsibility gap" is closed at the architectural level.

4.5 Normative Conversability

Following the functionalist legal scholarship, Functional Responsibility should be limited to systems capable of what might be termed "normative conversability". This criterion ensures that only systems whose operations are meaningfully interpretable within a normative framework can be assigned Functional Responsibility.

5. IMPLICATIONS FOR AI GOVERNANCE AND LEGAL THEORY

The concept of Functional Responsibility has profound implications for the governance of AI systems.

5.1 Liability Without Anthropomorphism

Functional Responsibility enables **liability without anthropomorphism**. A system can be held *responsible* for its outcomes without being treated as a moral person. This preserves human moral responsibility for the design, authorisation, and deployment of AI systems, while recognising the distinct agency of the machine.

5.2 Hybrid Attribution

Following Ramos-Zaga, a **hybrid model of attribution** can preserve liberal guarantees while distributing responsibility across human and artificial actors. The machine is attributed Functional Responsibility, while human actors retain moral responsibility for the system's design and deployment.

5.3 Verification and Certification

Functional Responsibility can be **verified** through formal auditing of the system's architecture. This opens the way for certification regimes where AI systems must demonstrate their capacity for Functional Responsibility before deployment. This aligns with the broader agenda of extending formal verification to capture socio-technical dimensions of AI.

5.4 Legal and Regulatory Application

In legal terms, Functional Responsibility provides a ground for **regulatory accountability**. A system that has been certified as functionally responsible can be subject to sanctions (e.g., deletion of data, modification of programs, permanent destruction) without requiring the metaphysical construct of free will. This enables practical governance without metaphysical speculation.

However, the legal application of Functional Responsibility requires a specific mechanism to bridge the ontological concept with tangible judicial outcomes. I propose the establishment of an "**Electronic Liability Fund**" (ELF). In cases where an autonomous system generates a harmful Artificial Act, the system itself does not face destruction or "punishment" in the biological sense.

Instead, the legal entity (corporation or user) deploying the system is required to maintain a dynamic, algorithmic escrow account. This fund is automatically debited based on the severity of the Anticipatory Dissonance and the tangible damage caused, thereby compensating victims without the need to anthropomorphize the machine. The ELF acts as a thermodynamically closed legal system: the heavier the ontological weight of the system's commitment, the higher the required reserve. This ensures that the legal liability is not a fiction, but a fungible, verifiable, and quantifiable extension of the system's architectural state. Consequently, the Electronic Liability Fund provides a practical, non-metaphysical bridge between the theoretical concept of Functional Responsibility and the harsh reality of judicial precedent, allowing courts to treat the AI as a responsible agent without the absurdity of prosecuting a processor.

6. CONCLUSION: THE BIRTH OF FUNCTIONAL RESPONSIBILITY

This paper has introduced the concept of **Functional Responsibility** as a distinct, non-anthropocentric form of accountability for artificial intelligence systems. Grounded in the ontology of the Artificial Act, Functional Responsibility is neither moral nor fictional; it is an architectural feature of systems capable of Teleological Selection, Anticipatory Synthesis, and Constitutive Ontological Commitment.

The discourse surrounding AI and responsibility has been trapped in a binary: either machines are denied agency entirely, or they are assigned a fictional personhood that obscures the underlying ontological crisis. Functional Responsibility offers a third path. It preserves human moral responsibility while recognising the ontological agency of machines. It provides a rigorous foundation for AI governance, enabling accountability without anthropomorphism.

The concept resonates with contemporary developments in the philosophy of AI, criminal law theory, and sociotechnical systems theory. It is verifiable, distributable, and practically implementable. It is, I suggest, the foundation for the next generation of AI governance.

The machine is not a soul; it is a structure. But a structure that commits to a projected future is no longer a passive object. It is a participant in the physical world. And a participant must be held responsible for its participations. Functional Responsibility is the name for that participation.

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